

A Modular Architecture for Automatic Tajweed Assessment of Quranic Recitation

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Chapter 1

Conceptual Framework

The preceding chapter established the linguistic foundations of Tajweed, reviewed related work on automatic recitation assessment, and identified the main limitations of existing approaches. Building on that foundation, this chapter establishes the conceptual framework adopted in this thesis for automatic Tajweed assessment of Qur’anic recitation.

The central argument of the chapter is that Tajweed assessment is not a single monolithic classification problem. Qur’anic recitation errors are heterogeneous in both their linguistic meaning and their acoustic manifestation. Some errors concern the recited content itself, such as omission, insertion, or substitution of a canonical unit. Other errors occur when the canonical content is present, but the learner fails to realise the applicable Tajweed rule correctly, such as insufficient elongation, weak nasalisation, incorrect assimilation, or missing release. Because these error types are structurally different, they require different forms of evidence, different modelling strategies, and different feedback mechanisms.

For this reason, the framework developed in this chapter is modular, reference-conditioned, and diagnosis-oriented. It separates content verification from rule-level Tajweed analysis, uses an annotated Qur’anic reference to determine where rules apply, routes each rule-bearing segment to an appropriate specialist module, and aggregates the resulting judgments into a structured diagnosis. The system is therefore designed not only to detect whether a recitation is incorrect, but also to identify where the deviation occurred, what type of deviation it represents, which Tajweed rule is involved, and how this information can support corrective feedback.

A key element of the framework is the use of a machine-readable annotated Tajweed reference. This reference layer gives the system explicit knowledge of the canonical text and of the Tajweed rules attached to each relevant position. As a result, the assessment process does not depend on

the acoustic signal alone. The audio provides evidence of what the learner produced, while the annotated reference defines what should have been produced and which rule should be checked at each position. This combination is what enables the proposed system to move beyond global correctness labels toward localised, rule-aware, and pedagogically meaningful diagnosis.

The chapter is organised as follows. Section 1.1 defines the assessment problem, introduces the learner-centred motivation of the system, and derives the main design requirements. Section 1.2 presents the modular architecture and the overall information flow. Section 1.3 discusses feature representation choices for different analysis modules. Section 1.4 presents the modelling strategy by Tajweed rule category. Section 1.5 explains diagnosis aggregation and feedback generation. Finally, Section 1.6 summarises the chapter.

1.1 The Multi-Faceted Nature of Recitation Errors

Before describing the architecture, it is necessary to define the assessment problem in a precise and methodical way. Automatic Tajweed assessment is not limited to deciding whether an audio recording is globally correct or incorrect. A useful system must distinguish between different sources of error, localise them in the recitation, relate them to the canonical Qur’anic reference, and return a diagnosis that can be transformed into meaningful learner feedback.

Errors in Qur’anic recitation do not arise from a single source and do not appear in a single acoustic form. Some errors concern the textual content of the recitation itself. Others concern the execution of Tajweed rules on otherwise correct content. A comprehensive assessment system must therefore represent these different error types explicitly, analyse them at the appropriate level, and preserve enough information to explain the final judgment.

1.1.1 Motivation and Expected System Behaviour

From the learner’s perspective, the system must behave less like a simple classifier and more like a diagnostic tutor. When a verse is recited, the objective is not merely to return a binary label such as *correct* or *incorrect*. The system must answer a set of practical pedagogical questions:

- **Content validity:** Did the learner recite the intended canonical content?
- **Localisation:** If a deviation occurred, at which word, character, phoneme, or aligned segment did it occur?
- **Rule applicability:** Which Tajweed rules are expected at the corresponding canonical position?
- **Rule execution:** If the content was correctly produced, were the applicable Tajweed rules realised correctly?
- **Error identification:** If a rule was misapplied, which rule was involved and what type of acoustic evidence supports that decision?
- **Corrective guidance:** How can the detected error be converted into feedback that helps the learner improve the next recitation attempt?

These requirements imply that the target output of the system is a structured diagnosis rather than a single label. The system must identify what was recited, align it with what should have been recited, determine which Tajweed rules are active at each relevant position, and then evaluate the learner’s realisation of those rules.

This also imposes a strict decision order. Content verification has interpretive priority over rule analysis. If the learner omits, inserts, or substitutes the segment that should carry a given Tajweed rule, the system records a content-level error first and does not treat the missing or altered segment as a normal rule-realisation case. If the segment is present and aligned with the canonical reference, the corresponding Tajweed rule can then be evaluated by the relevant specialist module. This ordering prevents the system from producing misleading rule feedback when the underlying recited content itself is incorrect.

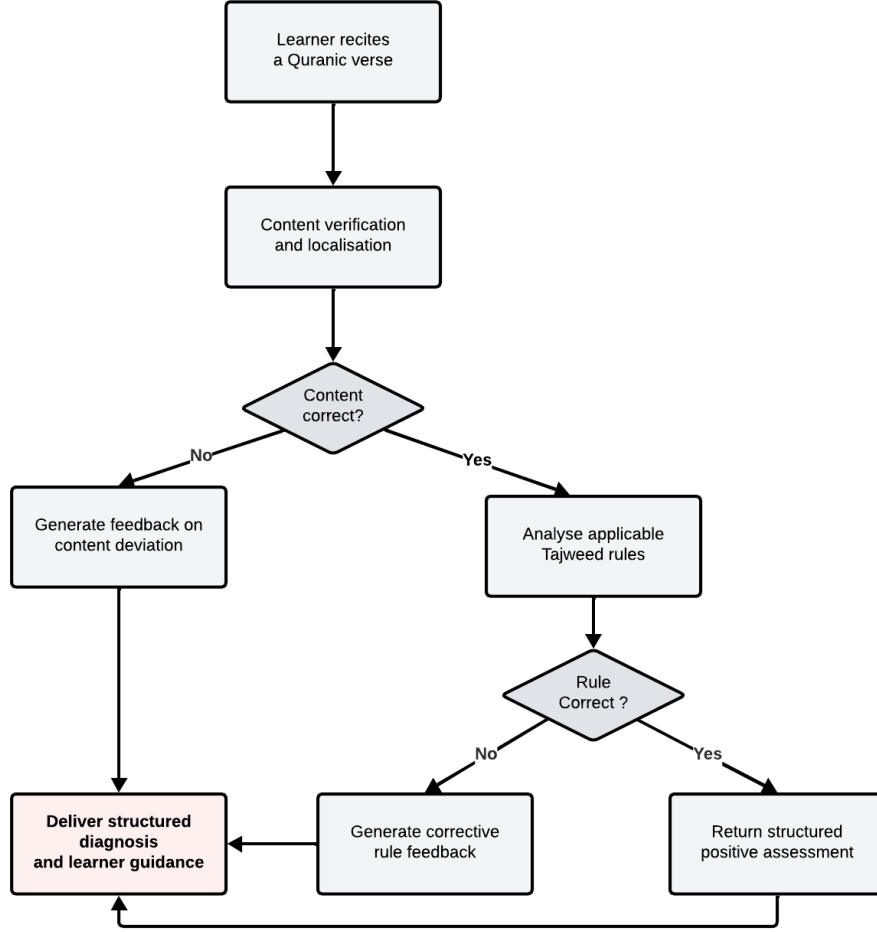


Figure 1.1: Learner-centred process flow of the recitation assessment system.

1.1.2 Formal Problem Formulation

Let the canonical recitation unit sequence be defined as

$$\mathcal{T} = (u_1, u_2, \dots, u_N), \quad (1.1)$$

where each u_i denotes a canonical unit of recitation at position i . Depending on the level of analysis, a unit may correspond to a character, phoneme, syllabic unit, word-level segment, or another alignment-compatible element of the canonical text.

Let the Tajweed rule inventory represented in the annotated Qur’anic reference be

$$\mathcal{R} = \{r_1, r_2, \dots, r_K\}. \quad (1.2)$$

The annotated reference defines a rule applicability function

$$\Phi : \{1, \dots, N\} \rightarrow 2^{\mathcal{R}}, \quad (1.3)$$

where $\Phi(i)$ denotes the set of Tajweed rules applicable at canonical position i . If no rule is active at position i , then $\Phi(i) = \emptyset$.

The same reference layer may also contain metadata associated with each rule-bearing position, such as the rule name, the textual span, the expected behaviour, and the feedback label. This can be represented abstractly as

$$\mathcal{K} = \left\{ (i, u_i, \Phi(i), \mu_i) \right\}_{i=1}^N, \quad (1.4)$$

where $\mathcal{K}$ denotes the machine-readable annotated Qur'anic knowledge layer and μ_i denotes the metadata attached to position i . This layer is central to the framework because it tells the system not only what the canonical unit is, but also which Tajweed rule should be expected at that location.

Let the learner's recorded recitation be represented by an acoustic sequence

$$\mathcal{X} = (x_1, x_2, \dots, x_T), \quad (1.5)$$

where each x_t denotes a feature vector extracted from the audio at frame t .

The first computational task is to determine which part of the learner's audio corresponds to which part of the canonical recitation. This alignment is represented as

$$\mathcal{A} = \{(i, s_i, e_i)\}_{i=1}^N, \quad (1.6)$$

where $[s_i, e_i]$ denotes the time interval in the learner's audio associated with canonical unit u_i , whenever such an interval can be reliably established. This alignment allows the system to attach each detected deviation to a precise canonical position instead of reporting only that an error occurred somewhere in the verse.

On the basis of the alignment, the system produces two main classes of judgments.

Content judgment. For each canonical position i , the system assigns a content status

$$c_i \in \{\text{correct, substitution, deletion, insertion}\}. \quad (1.7)$$

This judgment answers the question of whether the learner produced the intended canonical unit. The complete set of content judgments is

$$\mathbf{C} = \{c_i\}_{i=1}^N. \quad (1.8)$$

Rule judgment. For each rule $r \in \Phi(i)$ applicable at position i , the system assigns a rule status

$$z_{i,r} \in \{\text{correct, incorrect, not evaluated}\}. \quad (1.9)$$

The value *not evaluated* is used when the corresponding canonical segment is absent or unreliable because of a content-level error. This explicitly encodes the decision that rule analysis is meaningful only when the underlying content has been sufficiently verified.

To support explainability, each evaluated rule instance is associated with diagnostic evidence

$$\mathbf{d}_{i,r}, \quad (1.10)$$

whose interpretation depends on the rule category. For example, $\mathbf{d}_{i,r}$ may contain measured duration, nasality-related evidence, transition confidence, burst evidence, alignment quality, or the confidence score produced by a specialist module. If a rule is not evaluated, then $\mathbf{d}_{i,r}$ may be empty or undefined.

The decision policy for a rule-bearing position can therefore be summarised as

$$z_{i,r} = \begin{cases} \text{not evaluated,} & \text{if } c_i \neq \text{correct,} \\ h_{g(r)}(s_{i,r}), & \text{if } c_i = \text{correct and } r \in \Phi(i), \end{cases} \quad (1.11)$$

where $s_{i,r}$ is the aligned audio segment associated with rule r , $g(r)$ maps the rule to its specialist module category, and $h_{g(r)}$ denotes the corresponding rule-analysis function. This equation makes explicit that the system first verifies the content, then analyses Tajweed rules only on valid aligned segments.

The overall system can therefore be formalised as a mapping

$$f : (\mathcal{X}, \mathcal{T}, \mathcal{K}) \mapsto \mathcal{D}, \quad (1.12)$$

where $\mathcal{D}$ is a structured diagnosis object. A convenient abstract form for this output is

$$\mathcal{D} = \left\{ (i, u_i, c_i, \Phi(i), \mathcal{Z}_i, \mathcal{E}_i, \mathcal{F}_i) \right\}_{i=1}^N, \quad (1.13)$$

where $\mathcal{Z}_i$ denotes the set of rule-analysis outcomes at position i , $\mathcal{E}_i$ denotes the associated diagnostic evidence, and $\mathcal{F}_i$ denotes the learner-facing feedback generated for that position.

This formulation specifies what the system receives, what it knows, what it decides, and what it returns. It also clarifies the role of the annotated Qur’anic reference: the audio signal provides acoustic evidence, while the reference layer provides the canonical text, the applicable Tajweed rules, and the symbolic information needed for diagnosis and feedback.

1.1.3 A Functional Taxonomy of Recitation Errors

Under this formulation, recitation errors can be organised into three broad functional categories.

- **Content errors.** These occur when the learner deviates from the canonical text itself. They include word substitution, word omission, insertion, and phoneme substitution that changes lexical identity. For example, confusing قَلْب (qalb — heart) with كَلْب (kalb — dog) constitutes a content error with semantic consequences. Such errors are discrete and categorical: the learner either produced the intended unit or did not. Their detection depends on comparing *what was recited* with *what should have been recited*.
- **Tajweed rule misapplications.** These occur when the learner recites the intended content but does not realise the applicable Tajweed rule correctly. This category is internally heterogeneous and includes several acoustically distinct sub-types:
 - **Duration-based rules.** Madd and Ghunnah require the system to determine whether a sound has been sustained for the required interval. These are inherently temporal judgments.
 - **Transition-based rules.** Rules such as Idgham, Ikhfa', and Iqlab depend on how one sound evolves into the next. They require analysis of boundary dynamics rather than isolated static frames.
 - **Burst or release-based rules.** Rules such as Qalqalah depend on the presence of a brief, localised acoustic event associated with a specific consonantal release.
 - **Contextual realisation cases.** Some rules depend strongly on the surrounding canonical context and on the interaction between orthographic form and acoustic realisation. These cases require the system to reason not only about what is present in the signal, but also about what should be reduced, merged, concealed, or left inaudible.
- **Interacting errors.** In some cases, content errors and rule errors are closely related. For example, if a Madd-bearing segment is omitted altogether, the primary diagnosis is a content error because the expected segment itself is missing. By contrast, if the segment is present but

the elongation is too short or otherwise misrealised, the diagnosis becomes a rule-level error. This interaction is one of the main reasons the assessment problem cannot be handled reliably through a single undifferentiated classifier.

1.1.4 Why a Monolithic Approach is Insufficient

A single model trained to output one binary label for an entire recitation cannot provide a trustworthy diagnosis. At best, it can indicate that something is wrong. It cannot determine whether the problem concerns textual content, duration, transition, articulation, contextual realisation, or more than one category at the same time.

More fundamentally, the error categories described above require different representations and different modelling assumptions:

- **Content errors** require recognition and alignment mechanisms that estimate *what* was recited and compare it with the canonical sequence.
- **Duration-based rules** require models capable of tracking how long an acoustic state persists.
- **Transition-based rules** require contextual modelling across adjacent sounds and boundary regions.
- **Burst-based rules** require high temporal resolution and sensitivity to short local events.
- **Contextual and interacting cases** require explicit reasoning about whether the observed problem comes from incorrect content or from incorrect rule execution on valid content.

These are not minor implementation details. They correspond to different computational objectives, different forms of acoustic evidence, and different output semantics. A model specialised for recognising phoneme substitutions is not, by itself, the most appropriate mechanism for measuring elongation. Similarly, a model tuned for sustained duration is not naturally suited to detecting a brief Qalqalah-like release. Forcing all such cases into a single undifferentiated decision layer would obscure the diagnostic logic of the system and weaken the pedagogical value of the final output.

1.1.5 Requirements for a Comprehensive System

The preceding analysis leads directly to the following system requirements, each of which is addressed by a specific component of the architecture described in this chapter:

- **Reference-conditioned content verification.** The system compares the learner’s recitation against a canonical reference rather than operating in a reference-free manner.
- **Explicit localisation through alignment.** The system determines which part of the learner’s audio corresponds to which part of the canonical recitation, so that each detected error is attached to a precise textual location.
- **Rule-aware analysis.** The system knows which Tajweed rules are applicable at each canonical position and routes the corresponding acoustic evidence to the appropriate specialist module.
- **Specialised modelling.** The system supports different feature representations and modelling strategies for different acoustic categories of rules.
- **Structured multi-level output.** The system produces diagnostics at multiple granularities, including content status, rule status, localisation, and supporting acoustic evidence.
- **Explainable feedback generation.** The system transforms internal judgments into learner-facing feedback that identifies both the location and the nature of each error.

These requirements motivate the modular architecture presented in the following section. Rather than forcing all judgments through a single monolithic model, the design decomposes the assessment task into specialised but cooperating components whose outputs combine into a comprehensive and interpretable diagnosis.

1.2 Modular Architecture Overview

The preceding section established that Quranic recitation assessment is not a single homogeneous task. A complete system must distinguish between content-level deviations from the canonical text and rule-level deviations in Tajweed execution, while preserving the logical relation between them. This view is consistent with the limitations reported in prior work, where existing systems typically focus on a restricted subset of rules, specific error types, or isolated classification settings rather than a unified and pedagogically rich diagnosis [2, 3, 4, 6]. The architecture presented here addresses this gap through a modular design in which each component performs a clearly defined function and passes a structured output to the next stage.

Rather than treating the recitation as a single object to be classified globally, the system decomposes the assessment process into successive operations: feature extraction, content verification, alignment, rule-aware segment selection, specialised rule analysis, diagnosis aggregation, and feedback generation. This decomposition reflects both the heterogeneous acoustic nature of Tajweed-related errors and the broader finding in speech assessment research that different error categories benefit from different forms of modelling and diagnosis [1, 4, 18].

1.2.1 Architectural Principles

The architecture is guided by four design principles.

- **Separation of concerns.** Content verification is separated from rule analysis, and Tajweed rule analysis is itself divided into acoustically coherent sub-tasks. This prevents the system from treating all errors as if they had the same cause. It also responds to the fact that prior Qur’anic recitation systems commonly target limited rule families or narrow detection tasks, which shows the need for a framework that preserves task-specific structure rather than concealing it [2, 3, 6].
- **Reference awareness.** The system is explicitly conditioned on the canonical Qur’anic reference and its Tajweed annotations. It does not operate in a reference-free manner. Tajweed assessment is always judged against what should have been recited at a particular canonical position, not merely against what sounds plausible in isolation.
- **Structured intermediate representations.** Each stage produces an interpretable intermediate output, such as aligned intervals, content judgments, rule-specific evidence, or module confidence values. This

makes the assessment process traceable. It also supports diagnosis rather than mere detection, which aligns with the growing emphasis in mispronunciation research on fine-grained and explainable feedback rather than single-label correctness scoring [4, 18].

- **Pedagogical interpretability.** The final output must be meaningful for learners and teachers. For that reason, the architecture is designed not only to detect deviations, but also to preserve enough information to explain them. This directly addresses the pedagogical limitation of many automated recitation systems, which often stop at detection without providing sufficiently actionable corrective guidance [3, 6].

1.2.2 Formal Module Construction

Using the notation introduced in Section 1.1, the architecture can be described as a sequence of transformations over three main sources of information:

- **Acoustic input:** the learner’s recitation signal, represented after feature extraction as $\mathcal{X} = (x_1, x_2, \dots, x_T)$.
- **Canonical sequence:** the reference recitation unit sequence $\mathcal{T} = (u_1, u_2, \dots, u_N)$.
- **Annotated rule knowledge:** the machine-readable Tajweed reference layer $\mathcal{K}$, which defines the rule applicability function $\Phi(i)$ and the metadata associated with each rule-bearing position.

Let the set of functional modules be represented as

$$\mathcal{M} = \{m_{\text{feat}}, m_{\text{content}}, m_{\text{select}}, m_{\text{dur}}, m_{\text{trans}}, m_{\text{burst}}, m_{\text{ctx}}, m_{\text{agg}}, m_{\text{fb}}\}, \quad (1.14)$$

where the modules correspond respectively to feature extraction, content verification, rule-aware segment selection, duration analysis, transition analysis, burst analysis, contextual analysis, diagnosis aggregation, and feedback generation.

The feature extraction module converts the raw waveform into one or more frame-level representations:

$$m_{\text{feat}}(\text{audio}) \rightarrow \mathcal{X}. \quad (1.15)$$

The content verification and alignment stage then computes

$$m_{\text{content}}(\mathcal{X}, \mathcal{T}) \rightarrow (\hat{\mathcal{T}}, \mathcal{A}, \mathcal{C}), \quad (1.16)$$

where $\widehat{\mathcal{T}}$ is the recognised recitation hypothesis, $\mathcal{A}$ is the alignment between the learner’s audio and the canonical units, and $\mathcal{C} = \{c_i\}_{i=1}^N$ is the set of content judgments.

The rule-aware selection stage uses the alignment and the annotated reference to identify the rule-bearing segments:

$$m_{\text{select}}(\mathcal{A}, \Phi) \rightarrow \mathcal{S} = \{s_{i,r} \mid r \in \Phi(i)\}. \quad (1.17)$$

Here, $s_{i,r}$ denotes the audio segment associated with canonical position i and rule r .

The routing decision is controlled by a rule-to-module mapping

$$g : \mathcal{R} \rightarrow \{m_{\text{dur}}, m_{\text{trans}}, m_{\text{burst}}, m_{\text{ctx}}\}, \quad (1.18)$$

which assigns each Tajweed rule to the specialist module that best matches its dominant acoustic behaviour. For each valid rule instance (i, r) , the selected module produces

$$g(r)(s_{i,r}) \rightarrow (z_{i,r}, \mathbf{d}_{i,r}), \quad (1.19)$$

where $z_{i,r}$ is the rule judgment and $\mathbf{d}_{i,r}$ is the diagnostic evidence supporting that judgment.

Finally, the aggregation module combines content-level and rule-level outputs:

$$m_{\text{agg}}(\mathcal{C}, \{z_{i,r}, \mathbf{d}_{i,r}\}) \rightarrow \mathcal{D}, \quad (1.20)$$

where $\mathcal{D}$ is the structured diagnosis object. The feedback module then transforms this diagnosis into learner-facing guidance:

$$m_{\text{fb}}(\mathcal{D}, \mathcal{K}) \rightarrow \mathcal{F}. \quad (1.21)$$

This formal construction makes the architecture explicit. Each module has a defined input, a defined output, and a defined role in the complete assessment process. Such explicit modularisation is especially important in recitation assessment, where the system must support localisation, rule identification, evidence preservation, and feedback generation instead of only global classification [2, 4, 6].

1.2.3 Module Interfaces

Table 1.1 summarises the main components of the architecture and formalises their inputs and outputs.

Table 1.1: Main modules of the architecture with their inputs, outputs, and functions.

Module	Input	Output	Function
Feature extraction	Raw waveform	Frame-level feature sequence $\mathcal{X}$	Converts the recitation signal into numerical representations suitable for downstream analysis.
Content verification and alignment	$\mathcal{X}, \mathcal{T}$	$\hat{\mathcal{T}}, \mathcal{A}, \mathcal{C}$	Recognises what was recited, compares it with the canonical sequence, and localises content deviations.
Annotated rule reference	Canonical text and Tajweed annotation metadata	Rule map $\Phi(i)$ and reference knowledge layer $\mathcal{K}$	Specifies which Tajweed rules apply at each canonical position and supplies symbolic metadata for routing and feedback.
Rule-aware segment selection	$\mathcal{A}, \Phi$	Rule-linked audio segments $\mathcal{S}$	Determines which aligned audio intervals correspond to rule-bearing positions.
Specialised rule analysis modules	$s_{i,r}$	$z_{i,r}, \mathbf{d}_{i,r}$	Evaluates the execution of applicable Tajweed rules using rule-appropriate acoustic evidence.
Diagnosis aggregation	$\mathcal{C}, \{z_{i,r}, \mathbf{d}_{i,r}\}$	Structured diagnosis $\mathcal{D}$	Combines content and rule outputs into a coherent per-position diagnosis.
Feedback generation	$\mathcal{D}, \mathcal{K}$	Learner-facing feedback $\mathcal{F}$	Produces interpretable corrective guidance linked to the detected

1.2.4 Annotated Qur’anic Reference as a Rule Knowledge Layer

The system relies on an annotated Tajweed reference rather than on plain text alone. This reference identifies the textual spans to which Tajweed rules apply and associates each span with one or more rule labels. In the implemented framework, this role is naturally represented by a structured annotation format such as JSON. Conceptually, however, its function is broader than data storage: it acts as a rule knowledge layer that connects canonical text, Tajweed rules, expected acoustic behaviour, and feedback metadata.

An annotated rule instance can be represented abstractly as

$$\kappa_j = (\text{span}_j, r_j, c_j, \eta_j), \quad (1.22)$$

where span_j denotes the canonical textual span, r_j denotes the Tajweed rule label, c_j denotes the acoustic category of the rule, and η_j denotes additional metadata such as expected duration, rule description, or feedback wording. The complete annotated reference is then

$$\mathcal{K} = \{\kappa_1, \kappa_2, \dots, \kappa_L\}. \quad (1.23)$$

This layer provides four essential functions:

- **Locating rule-bearing positions.** It indicates where rule analysis is expected in the canonical recitation.
- **Specifying the applicable rule.** It tells the system which Tajweed rule should be checked at each annotated position.
- **Supporting rule-aware routing.** It allows the system to map a rule instance to the appropriate specialist module, such as a duration, transition, burst, or contextual module.
- **Supporting diagnosis and feedback.** It provides the symbolic rule metadata needed to express the final diagnosis in a pedagogically meaningful form.

This reference layer is essential because the audio signal reveals what the learner actually produced, whereas the annotated canonical resource determines what should have been realised at that position. The distinction is central to Tajweed assessment. It also makes the system more explainable: when a module detects an error, the system can connect that acoustic evidence back to the exact rule and location stored in the annotated reference.

Figure 1.2 illustrates the role of this annotated reference layer inside the architecture.

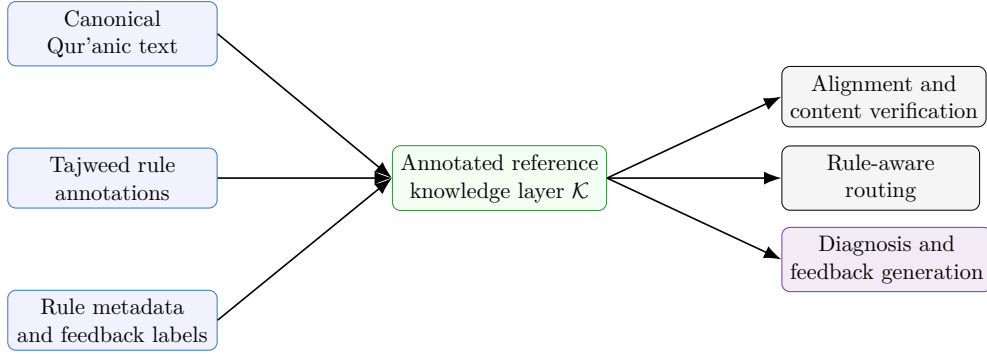


Figure 1.2: Role of the annotated Qur’anic reference as a rule knowledge layer. The reference combines canonical text, Tajweed rule annotations, and rule metadata, then supplies this information to alignment, routing, diagnosis, and feedback generation.

This annotated reference can be viewed as a lightweight knowledge graph for the purposes of the assessment system. It links textual units to rule labels, rule labels to acoustic categories, and detected errors to feedback descriptions. The system therefore does not need to infer the entire rule context from audio alone. Instead, it evaluates the learner’s audio in relation to an explicit symbolic model of the expected recitation.

This decomposition improves feedback quality. If the duration module signals an error, the system already knows that the problem belongs to a duration-related family, such as Madd or Ghunnah. If the transition module signals an error, the system narrows the problem to boundary phenomena such as Ikhfa’, Idgham, or Iqlab. The annotated reference then identifies the exact rule expected at the aligned position. The feedback layer can therefore generate a much more informative response than a generic statement such as “incorrect recitation”.

1.2.5 Overall System Flow

Figure 1.3 presents the overall logic of the architecture. The figure emphasises the main decision principle of the system: content verification precedes Tajweed rule analysis. If the recited content deviates from the canonical reference, the system records a content error first and sends this diagnosis to the aggregation and feedback stages. If the content is correctly aligned, the

system retrieves the applicable Tajweed rules and routes the corresponding audio segment to the appropriate specialist module.

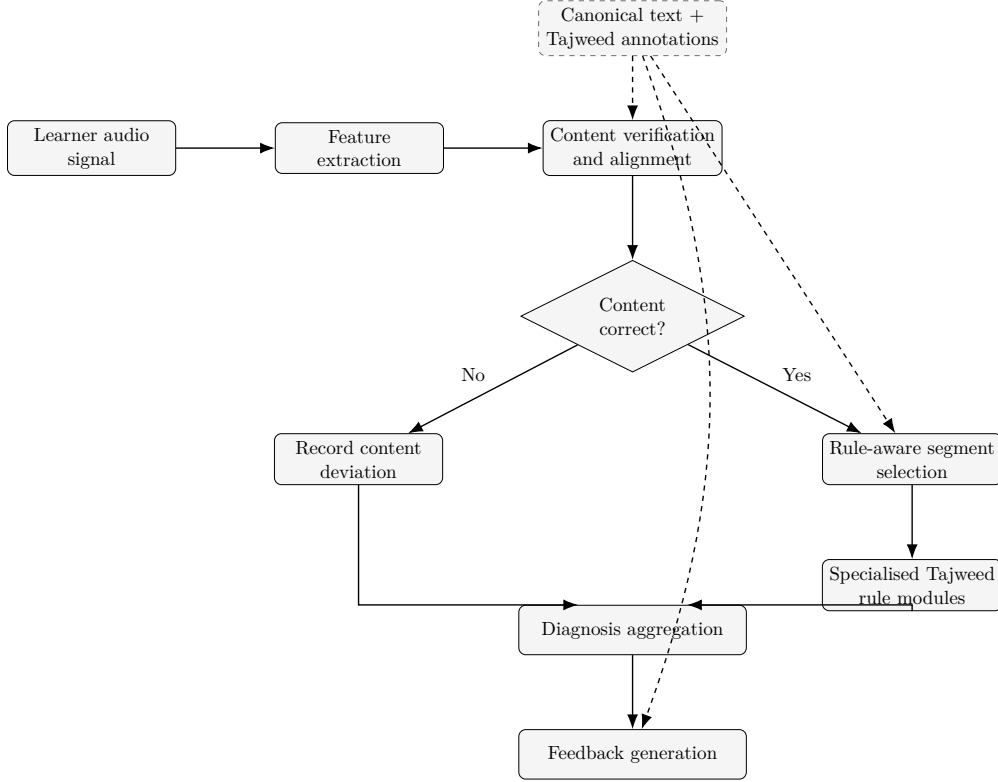


Figure 1.3: Detailed system flow of the modular architecture. Dashed arrows denote the use of canonical Qur’anic and Tajweed reference metadata.

The same flow can be expressed as the following decision procedure:

1. Receive the learner’s audio signal and extract the required acoustic representations.
2. Align the learner’s recitation with the canonical Qur’anic sequence.
3. For each canonical position i , compute the content judgment c_i .
4. If $c_i \neq \text{correct}$, record a content-level deviation and skip Tajweed rule analysis for that position.
5. If $c_i = \text{correct}$, retrieve the applicable rule set $\Phi(i)$ from the annotated reference.

6. For each rule $r \in \Phi(i)$, use the routing function $g(r)$ to select the appropriate specialist module.
7. Apply the selected module to the aligned segment $s_{i,r}$ and produce the rule judgment $z_{i,r}$ with diagnostic evidence $\mathbf{d}_{i,r}$.
8. Aggregate content-level and rule-level judgments into the structured diagnosis $\mathcal{D}$.
9. Generate learner-facing feedback from $\mathcal{D}$ and the metadata stored in the annotated reference layer $\mathcal{K}$.

This procedure makes the conditional logic of the system explicit. Rule analysis is not performed blindly on every audio segment. It is performed only when the content has been verified and when the annotated reference indicates that a rule is expected at the corresponding position.

1.2.6 Specialised Rule Analysis

A key architectural choice is that Tajweed rule analysis is not implemented as one undifferentiated classifier. Instead, the system groups rule instances by acoustic behaviour and routes them to specialised modules. This makes the design faithful to the phonetic reality of Tajweed and strengthens the interpretability of the final diagnosis.

For the purposes of system design, the rule inventory is grouped into the following broad categories:

- **Duration-oriented rules.** These include elongation and sustained nasalisation phenomena, such as Madd variants and Ghunnah. Their analysis depends primarily on temporal persistence and duration-sensitive acoustic evidence. Prior work on Qur’anic rule recognition and speech assessment supports the importance of temporal modelling for such phenomena [2, 3, 1].
- **Transition-oriented rules.** These include concealment, assimilation, and related boundary phenomena, such as Ikhfa’, Iqlab, and the different forms of Idgham. Their analysis depends on how one sound evolves into the next across a local phonetic transition.
- **Release- or event-oriented rules.** These include rules such as Qalqalah, where a short local acoustic event is central to the judgment.

- **Contextual realisation rules.** These include cases whose behaviour depends on textual context, such as Hamzat al-Wasl, Lam Shamsiyyah, and silent segments. These rules are strongly conditioned by the canonical reference and by the interaction between orthographic and acoustic realisation.

This organisation does not imply that every rule has only one acoustic property. Some rules may involve more than one cue. The purpose of the categorisation is architectural: it defines how rule instances are routed, which type of detector is responsible for them, and how the resulting evidence should be interpreted. It also reflects the broader trend in the literature toward fine-grained rather than purely monolithic pronunciation diagnosis [4, 18].

1.2.7 Value Added by Each Architectural Component

The value of the modular architecture lies in the contribution of each component to the final diagnosis. Table 1.2 summarises the specific role played by each major component.

Table 1.2: Diagnostic value added by the main architectural components.

Component	Value added to the assessment
Content verification	Prevents the system from analysing Tajweed rules on missing, substituted, or incorrectly aligned content. It establishes what was actually recited before rule-level judgment is attempted.
Alignment	Localises the learner’s audio with respect to the canonical text, allowing errors to be attached to precise positions instead of being reported only at verse level.
Annotated reference layer	Supplies explicit knowledge of which rule is expected at each canonical position, enabling rule-aware routing and rule-specific diagnosis.
Duration module	Narrows the diagnostic space to temporal errors such as Madd and Ghunnah, where the relevant evidence is persistence over time.
Transition module	Handles boundary phenomena such as Ikhfa’, Idgham, and Iqlab, where the relevant evidence lies in how one sound evolves into the next.
Burst module	Detects short local release events such as Qalqalah, which require high temporal sensitivity and cannot be treated as ordinary long duration phenomena.
Contextual module	Handles cases whose interpretation depends strongly on the canonical context and on whether a sound should be pronounced, merged, reduced, or suppressed.
Aggregation layer	Combines content-level and rule-level judgments into a coherent diagnosis while preserving evidence and avoiding contradictory outputs.
Feedback layer	Converts the structured diagnosis into learner-facing guidance that identifies the location, rule, error type, and correction path.

1.2.8 Extensibility of the Architecture

The architecture is extensible by design. Because the reference layer explicitly stores rule tags and the rule analysis stage is modular, new Tajweed rules can be incorporated in one of two ways:

- **Reuse an existing module.** A new rule can be assigned to an existing acoustic module if its behaviour matches an already modelled category.

- **Introduce a new specialist module.** A new detector can be added if the rule requires a distinct analysis strategy.

This extensibility is particularly important for Qur’anic recitation assessment, where rule inventories may vary by annotation scheme, pedagogical focus, or recitation style. A modular design therefore offers a more durable foundation than a single tightly coupled classifier. It allows the system to expand rule coverage while preserving the same alignment, routing, aggregation, and feedback logic. This is especially valuable in light of the diversity of task formulations already observed in the literature [2, 3, 6].

1.3 Feature Representation

Before any linguistic or rule-level analysis can be performed, the raw recitation signal must be transformed into a representation that a model can process. Feature extraction is therefore not a secondary preprocessing step; it directly determines which acoustic information remains available to the downstream modules. In Tajweed assessment, this is particularly important because different types of errors depend on different acoustic cues. A duration error, a transition error, a brief release error, and a content-recognition error do not require exactly the same representation.

For this reason, the architecture adopts a module-dependent feature strategy. Rather than imposing one feature family on the whole system, the framework uses the representation that best matches the function of each module. Hand-crafted acoustic features are used where local spectral and temporal evidence is central, while self-supervised speech representations are used where robust content recognition and alignment are required. This design follows directly from the modular architecture: because the system separates content verification, duration analysis, transition analysis, and burst analysis, the feature layer must also support these different analytical needs.

1.3.1 Functional Role of Features in Tajweed Assessment

In the context of Tajweed assessment, the feature layer supports three main forms of analysis.

- **Content verification.** The system must identify what was recited and compare it with the canonical Qur’anic sequence. This requires representations that preserve phonetic contrasts and remain robust across speakers, recording conditions, and recitation styles.
- **Temporal analysis.** Rules such as Madd and Ghunnah depend on how long a sound is sustained. This requires frame-level representations that preserve temporal continuity and allow the model to observe persistence across consecutive frames.
- **Rule-specific acoustic analysis.** Rules such as Ikhfa’, Idgham, Iqlab, and Qalqalah depend on local acoustic details, transitions between sounds, or short release events. This requires representations that preserve fine-grained time-frequency information while remaining suitable for module-level classification.

These requirements show that no single feature family is equally optimal for all parts of the system. Classical descriptors such as MFCCs remain useful for compact local spectral analysis, while self-supervised speech representations provide richer phonetic and contextual information [8, 17, 5, 13, 15]. The adopted feature strategy therefore combines both families in a controlled and module-specific manner.

1.3.2 Feature Extraction Pipeline

Figure 1.4 summarises the feature extraction strategy used in the framework. The same raw recitation waveform can be processed through two complementary branches. The first branch extracts hand-crafted MFCC-based features, including first- and second-order temporal derivatives when required. The second branch extracts learned speech representations using a self-supervised encoder. The downstream modules then use the branch that matches their diagnostic role.

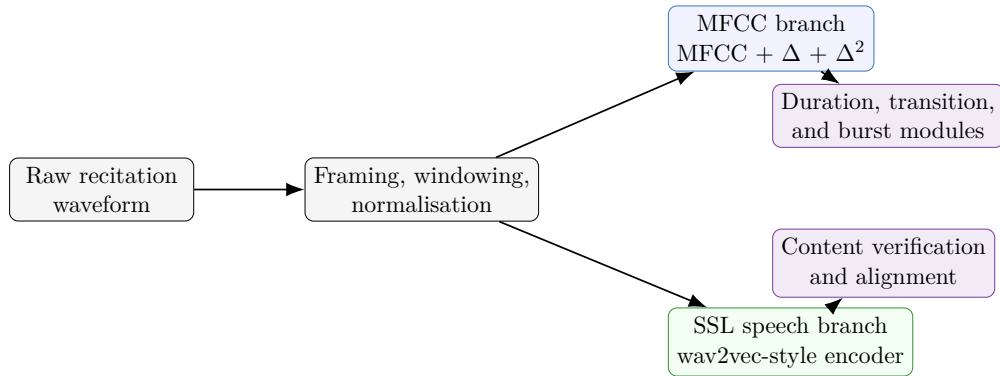


Figure 1.4: Feature extraction pipeline of the modular Tajweed assessment system. MFCC-based features provide local frame-level evidence for rule-oriented modules, while self-supervised speech representations support content verification and alignment.

This figure reflects the logic of the architecture. The MFCC branch is not used merely because it is traditional; it is used because duration, burst, and local transition phenomena require compact frame-level acoustic evidence. Similarly, the self-supervised branch is not used as a general replacement for all features; it is assigned to the content path because content verification is closest to speech recognition and therefore benefits from stronger learned speech representations.

1.3.3 Hand-Crafted Acoustic Features

Among hand-crafted speech features, MFCCs remain the most relevant to the present system. They are extracted by framing the signal, computing the short-time spectrum, projecting it onto the Mel scale, applying logarithmic compression, and using a discrete cosine transform to obtain a compact cepstral representation [8]. First-order delta coefficients and second-order delta-delta coefficients can then be added to represent short-term dynamic change [17].

In this framework, the MFCC branch produces a frame-level sequence of the form

$$\mathbf{m}_t = [\text{MFCC}_t, \Delta\text{MFCC}_t, \Delta^2\text{MFCC}_t], \quad (1.24)$$

where $\mathbf{m}_t$ denotes the acoustic feature vector at frame t . The full MFCC-based sequence is therefore

$$\mathcal{X}_{\text{MFCC}} = (\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_T). \quad (1.25)$$

MFCC-based representations remain widely used in Tajweed-oriented and Arabic mispronunciation systems. Al-Ayyoub et al. used MFCCs as part of a larger feature combination for Tajweed rule classification [2]. Al Harere and Al Jallad used MFCC features with LSTM models for basic Qur’anic recitation rule mispronunciation detection [3]. Ahmed et al. also reported strong use of MFCC-style features in Arabic mispronunciation detection with LSTM networks [1]. More recent Tajweed classification work by Salim et al. likewise uses MFCC representations with CNN-based classification [16].

Their continued use is justified by several practical advantages:

- **Low computational cost.** MFCCs can be extracted efficiently and do not require large-scale pre-training.
- **Good local spectral representation.** They preserve information related to vocal-tract shape, phonetic contrast, and short local acoustic events.
- **Compatibility with temporal modelling.** Because they are extracted frame by frame, they are naturally suitable for models that analyse temporal persistence, such as duration modules.
- **Useful dynamic information.** Delta and delta-delta coefficients provide additional evidence about short-term acoustic change, which is relevant for transitions and localised rule phenomena.

- **Relative interpretability.** MFCCs remain easier to analyse and justify than high-dimensional learned embeddings.

For these reasons, the MFCC branch is the natural representation for duration-oriented modules such as Madd and Ghunnah, and it is also appropriate for local rule modules that depend on short acoustic events or local transitions.

1.3.4 Self-Supervised Speech Representations

The second feature family consists of self-supervised speech representations, such as those produced by wav2vec 2.0, HuBERT, Whisper-style encoders, and related audio-oriented Transformer models [5, 13, 15]. These models are pre-trained on large speech corpora and learn representations that capture not only local acoustic patterns, but also higher-level phonetic and contextual information.

In the proposed framework, this branch is assigned primarily to content verification and alignment. This is because the content path must estimate what the learner recited, compare it with the canonical sequence, and support alignment between the learner’s audio and the reference. These tasks are closer to speech recognition and phoneme-level mispronunciation detection than to isolated Tajweed rule classification. They therefore benefit from learned speech representations that encode broader phonetic context.

Recent Arabic and Qur’anic pronunciation-assessment studies support this choice. Alrashoudi et al. used Transformer-based speech models, including wav2vec 2.0, HuBERT, and Whisper variants, for Arabic phoneme-level mispronunciation detection and diagnosis [4]. Çalık et al. evaluated audio-oriented Transformer models, including SEW, HuBERT, wav2vec, and UniSpeech, for Arabic phoneme mispronunciation detection [7]. More recently, El Kheir et al. introduced QuranMB.v1 as a unified benchmark for Arabic pronunciation assessment using Qur’anic recitation as a case study [9]. In the same research direction, the Hafs2Vec system used a wav2vec2-XLS-R model together with a Qur’anic phonemizer for Arabic and Qur’anic phoneme-level pronunciation assessment [14].

Compared with hand-crafted features, self-supervised speech representations offer several advantages:

- **Contextual richness.** The representation of a frame or segment is informed by broader surrounding context, not only by its local spectral shape.

- **Robustness.** Large-scale pre-training improves tolerance to speaker variation, recording conditions, and pronunciation variability.
- **Transferability.** Pre-trained encoders can be adapted to Arabic or Qur’anic recitation tasks without requiring a very large task-specific corpus from the beginning.
- **Suitability for recognition.** Recent Arabic and Qur’anic pronunciation-assessment work shows the strength of self-supervised and Transformer-based speech models for recognition-oriented tasks, including phoneme recognition, mispronunciation detection, and pronunciation assessment [4, 7, 9, 14].

Thus, the SSL branch supports the part of the system where robust recognition and alignment are most important. It complements rather than replaces the MFCC branch.

1.3.5 What Related Work Suggests About Feature Choice

The reviewed literature does not support a single universal feature choice for all parts of a Tajweed assessment system. Instead, it shows a clear division between the strengths of hand-crafted acoustic features and learned speech representations.

Tajweed-specific systems have relied heavily on MFCC-style features. Al Harere and Al Jallad used MFCCs with LSTM networks and achieved strong results on basic Tajweed rule mispronunciation detection [3]. Al-Ayyoub et al. combined MFCCs with other hand-crafted and learned features for multi-class Tajweed rule classification [2]. Salim et al. used MFCCs with CNN models for Tajweed law recognition [16]. These works indicate that MFCC-based representations remain a strong and defensible choice for rule-level acoustic modelling, especially when the target phenomenon is local, temporal, or spectrally distinct.

At the same time, more recent Arabic and Qur’anic pronunciation assessment work shows the value of self-supervised speech encoders. Alrashoudi et al. report strong performance from Transformer-based models such as wav2vec 2.0, HuBERT, and Whisper variants for Arabic phoneme-level mispronunciation detection and diagnosis [4]. Çalik et al. show that audio-oriented Transformer models can be applied effectively to Arabic phoneme mispronunciation detection [7]. El Kheir et al. and Ibrahim provide more recent Qur’anic pronunciation-assessment references, showing that this direction is also becoming relevant specifically for Qur’anic recitation [9, 14].

This supports the use of self-supervised representations for the content verification path, where the system must recover what was recited and align it with the canonical reference.

Finally, fusion-based work in Qur’anic recitation classification shows that complementary feature streams can be useful when the task requires both local acoustic precision and broader contextual information [11]. This does not mean that every module should use all available features. Rather, it supports the principle that feature choice should be driven by the diagnostic role of each module.

The resulting design choice is therefore deliberate:

- MFCC-based features are used for local and temporal rule-oriented modules.
- Self-supervised speech representations are used for content verification and alignment.
- Hybrid use remains conceptually available for modules whose target phenomena require both local acoustic precision and broader contextual interpretation.

1.3.6 Comparative View of the Two Feature Families

Table 1.3 summarises the main differences between the two feature families across criteria that matter for the present system.

Table 1.3: Comparison of hand-crafted and self-supervised feature families for Tajweed assessment.

Aspect	Hand-crafted features (MFCC family)	Self-supervised speech features
Main strength	Compact local spectral and temporal representation	Rich phonetic and contextual representation
Typical use in related work	Tajweed rule classification and Arabic mispronunciation detection using MFCC, MFCC-LSTM, or MFCC-CNN pipelines	Arabic and Qur’anic phoneme recognition, mispronunciation detection, and pronunciation assessment using wav2vec-style, HuBERT, Whisper-style, and related Transformer models
Data requirement	No training required for feature extraction	Requires large-scale pre-training, but can be adapted through fine-tuning or feature extraction
Computational cost	Low	Higher
Interpretability	Relatively high	Lower, because the representation is learned and high-dimensional
Suitability for content verification	Moderate	High
Suitability for duration-based rules	High, especially with delta and delta-delta coefficients	Moderate to high, but less directly interpretable for duration measurement
Suitability for local event detection	High for short acoustic events such as releases or bursts	Moderate, depending on model resolution and downstream classifier
Suitability for context-dependent rules	Moderate	High
Best use in this framework	Duration, transition-support, and burst-oriented modules	Content verification and alignment

1.3.7 Feature Strategy by Module

The feature strategy adopted in the framework is therefore module-specific.

- **Content verification and alignment.** This module uses self-supervised speech representations because its main task is to recover the recited content and align it with the canonical sequence. This choice is supported by recent Arabic and Qur’anic pronunciation-assessment work showing the strength of self-supervised and Transformer-based models for recognition-oriented tasks [4, 7, 9, 14].
- **Duration-oriented rule modules.** Madd and Ghunnah are analysed using MFCC-based frame-level representations enhanced with delta and delta-delta coefficients. This representation is appropriate because the judgment depends on persistence over time and on measurable acoustic continuity.
- **Transition-oriented rule modules.** Rules such as Ikhfa’, Idgham, and Iqlab require sensitivity to how one sound changes into the next. MFCC-based frame sequences provide local acoustic evidence for this boundary analysis, while the architecture remains compatible with contextual representations when broader phonetic context is needed.
- **Burst-oriented rule modules.** Qalqalah and similar release-based phenomena require attention to short, local acoustic events. MFCC-based local representations are suitable for this module because they preserve the time-frequency detail needed to detect brief releases.
- **Contextual-realisation cases.** Rules whose interpretation depends strongly on the canonical context can combine aligned acoustic evidence with information from the annotated reference. In these cases, the decisive factor is not only the acoustic feature vector, but also the rule label and textual context supplied by the reference layer.

This strategy makes the feature layer consistent with the overall architecture. The system does not select features independently from the modelling problem. Instead, each representation is assigned to the module for which it provides the most useful evidence: self-supervised features for content recognition and alignment, and MFCC-based features for local, temporal, and rule-oriented acoustic analysis.

1.4 Modelling Strategy by Rule Category

The modelling strategy follows directly from the modular view of Tajweed assessment developed in the previous sections. The main question is not which single global architecture should be used for all errors, but how each category of Tajweed rule should be modelled according to its acoustic and linguistic behaviour. Some rules depend mainly on duration, others on transitions between sounds, others on short local release events, and others on the interaction between the acoustic signal and the canonical textual context. A single uniform classifier cannot represent these differences in a transparent or pedagogically useful way.

For this reason, the system adopts a rule-aware modelling strategy. After content verification and alignment, the system does not analyse the whole recitation as one undifferentiated signal. Instead, it analyses validated aligned segments that are already linked to canonical positions and to the Tajweed rules expected at those positions. The annotated reference supplies the rule tags, the routing layer maps those tags to modelling categories, and each specialist module evaluates the type of acoustic evidence relevant to its category.

This modelling strategy is consistent with prior work in Qur’anic recitation and Arabic pronunciation assessment, where different model families have been used depending on the target phenomenon, available data, and task granularity [2, 3, 1, 4, 16]. It also gives the proposed framework a stronger diagnostic value: when a module detects an error, the system already knows the broad nature of the error before feedback is generated.

1.4.1 Why Modelling Must Follow Rule Categories

At this stage of the pipeline, the input is no longer raw audio. The system operates on an aligned segment that has already been connected to a canonical position. The modelling problem is therefore defined by three pieces of information:

- the aligned audio segment associated with the canonical position;
- the Tajweed rule or rules retrieved from the annotated reference;
- the acoustic category to which each rule belongs.

To formalise this, let

$$\mathcal{Q} = \{q_{\text{dur}}, q_{\text{trans}}, q_{\text{burst}}, q_{\text{ctx}}\} \quad (1.26)$$

be the set of modelling categories, corresponding respectively to duration-oriented, transition-oriented, burst-oriented, and contextual-realisation rules.

Let

$$\gamma : \mathcal{R} \rightarrow \mathcal{Q} \quad (1.27)$$

be the rule-to-category mapping function, where $\mathcal{R}$ is the set of Tajweed rules represented in the annotated reference. The notation $\gamma(r)$ therefore identifies the modelling category associated with rule r .

Let the set of specialist rule modules be

$$\mathcal{H} = \{h_{\text{dur}}, h_{\text{trans}}, h_{\text{burst}}, h_{\text{ctx}}\}, \quad (1.28)$$

where each h corresponds to the detector or classifier responsible for one rule category. A routing function

$$\rho : \mathcal{Q} \rightarrow \mathcal{H} \quad (1.29)$$

then assigns each modelling category to its corresponding specialist module.

For a canonical position i with applicable rule set $\Phi(i)$, the set of responsible modelling categories is

$$\mathcal{Q}(i) = \{\gamma(r) \mid r \in \Phi(i)\}. \quad (1.30)$$

The corresponding set of responsible specialist modules is

$$\mathcal{H}(i) = \{\rho(\gamma(r)) \mid r \in \Phi(i)\}. \quad (1.31)$$

For each rule instance (i, r) , the selected specialist module produces a rule judgment and supporting evidence:

$$\rho(\gamma(r))(s_{i,r}) \rightarrow (z_{i,r}, \mathbf{d}_{i,r}), \quad (1.32)$$

where $s_{i,r}$ is the aligned audio segment associated with rule r , $z_{i,r}$ is the rule-level judgment, and $\mathbf{d}_{i,r}$ is the diagnostic evidence produced by the selected module.

This formulation makes the routing logic explicit. The responsible model is not chosen arbitrarily from the audio signal alone. It is determined by the combination of alignment, annotated rule tags, rule-category mapping, and specialist module routing. If more than one Tajweed rule applies within the same broader segment, the system creates multiple rule instances and routes each one to the appropriate module before aggregation.

1.4.2 Rule-Guided Modelling Pipeline

Figure 1.5 summarises the rule-guided modelling process. The input to the modelling stage is a validated aligned segment, not the whole recitation. The system retrieves the applicable rule set $\Phi(i)$, maps each rule to an acoustic category, and routes the segment to the corresponding specialist module.

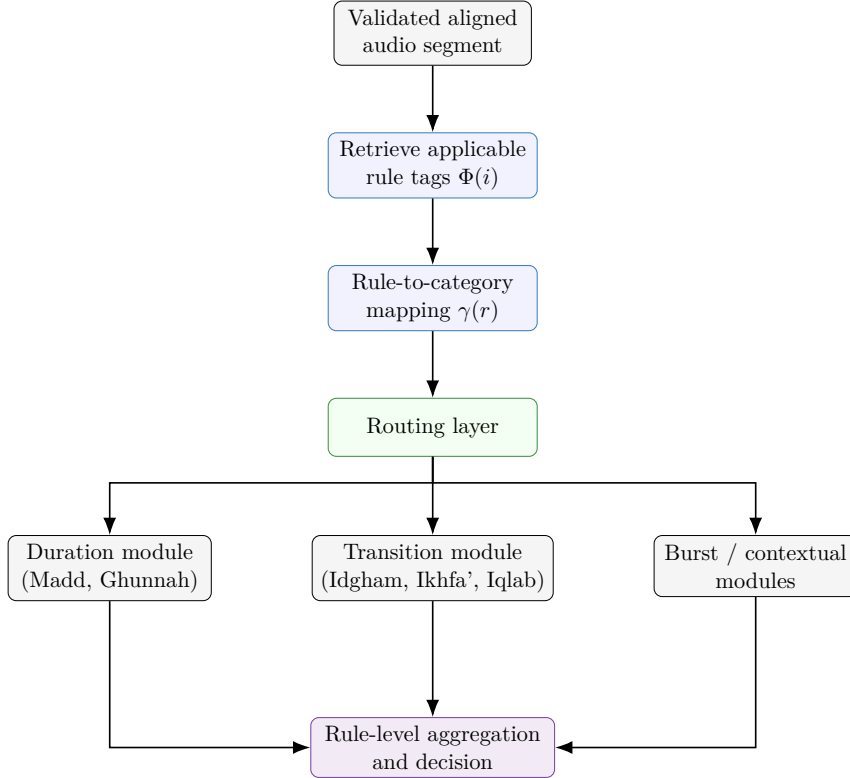


Figure 1.5: Rule-guided modelling pipeline. A validated aligned segment is linked to its Tajweed rule tags, mapped to acoustic modelling categories, and routed to one or more specialist modules before rule-level aggregation.

This routing mechanism answers a practical modelling question: how does the system know which model should analyse a given segment? The answer is that responsibility is determined by the annotated reference and the rule-to-category mapping. The model is therefore not asked to infer the full rule context from audio alone. It evaluates an aligned segment in light of the rule that is known to be expected at that canonical position.

1.4.3 Specialisation by Acoustic Category

The modelling layer becomes more interpretable when the Tajweed rule inventory is grouped according to dominant acoustic behaviour. For the present framework, four broad modelling categories are used.

- **Duration-oriented rules.** These include rules such as Madd and Ghunnah. The main evidence is the persistence of a sound over time. The module must therefore determine whether the relevant acoustic segment has been sustained for the expected interval.
- **Transition-oriented rules.** These include Idgham, Ikhfa', and Iqlab. The main evidence lies in how one sound changes into the next. The module must therefore analyse local boundary dynamics rather than a single isolated frame.
- **Burst-oriented rules.** These include Qalqalah and similar cases in which a short local acoustic event is central to the judgment. The module must be sensitive to brief release-like cues.
- **Contextual-realisation rules.** These include cases such as Hamzat al-Wasl, Lam Shamsiyyah, and silent or reduced segments. The correct decision depends strongly on canonical context and on whether the expected sound should be pronounced, merged, reduced, or suppressed.

This categorisation does not imply that every Tajweed rule has only one acoustic property. Some rules combine duration, transition, and contextual cues. The purpose of the categorisation is architectural: it defines which module is responsible for a rule instance and how the resulting evidence should be interpreted.

1.4.4 Rule Category Responsibilities

Table 1.4 summarises the responsibility of each modelling category in terms of rule examples, dominant evidence, and expected module output.

Table 1.4: Rule categories, dominant acoustic evidence, and module outputs.

Category	Example rules	Dominant evidence	Module output
Duration-oriented	Madd, Ghunnah	Persistence of the relevant sound across time; duration-sensitive frame-level evidence.	Rule judgment with duration-related evidence, such as whether the sound was sufficiently sustained.
Transition-oriented	Ikhfa', Idgham, Iqlab	Boundary dynamics between adjacent sounds; gradual concealment, assimilation, or conversion.	Rule judgment with transition confidence or local boundary evidence.
Burst-oriented	Qalqalah	Short local acoustic release or burst-like event near the target consonant.	Rule judgment with local event evidence.
Contextual-realisation	Hamzat al-Wasl, Lam Shamsiyyah, silent or reduced segments	Interaction between canonical text, rule context, and observed acoustic realisation.	Rule judgment conditioned on the aligned acoustic segment and the annotated reference context.

This table clarifies the value of the rule-category strategy. Each category defines what the system should look for, which type of model is appropriate, and what kind of evidence should be preserved for later aggregation and feedback.

1.4.5 Internal Structure of the Rule Analysis Stage

Figure 1.6 refines the previous pipeline by showing the internal organisation of the modular rule-analysis stage. The key element is the routing layer. The aligned segment is not simply copied to all possible models. Instead,

the routing stage uses the applicable rule tags and the rule-category lookup table to determine which specialist module or modules should process the segment.

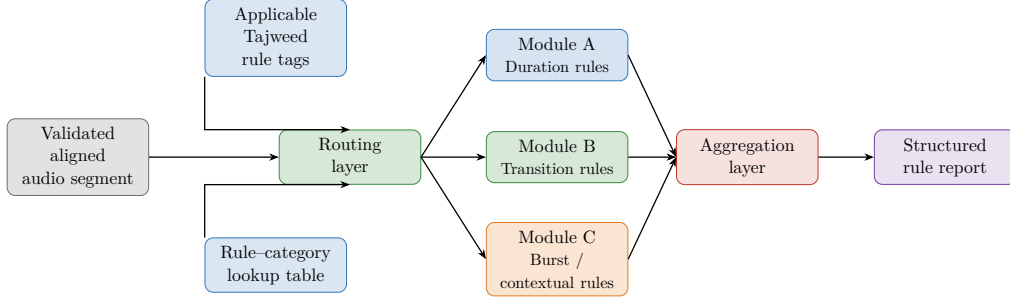


Figure 1.6: Internal structure of the rule-analysis stage. The routing layer determines which specialist module is responsible for a validated segment by combining rule tags from the annotated reference with the rule-to-category mapping. A segment may be sent to more than one module when multiple rule categories apply.

This figure also clarifies the unit of analysis. The routed unit is not necessarily an entire *ayah*. It is the aligned segment associated with the canonical position under analysis. If several Tajweed rules are relevant within the same broader Qur’anic segment, the system creates separate rule instances and routes them according to their categories.

1.4.6 Baseline Model Choice by Category

Once the routing logic is established, each category can be assigned a modelling strategy that matches its dominant evidence type. The framework adopts the following category-specific baseline choices.

- **Duration-oriented rules: MFCC-based sequential model with local duration evidence.** Madd and Ghunnah require the system to track persistence over time. MFCC-based frame sequences provide compact acoustic evidence, while a sequential model captures the temporal continuity of the target segment. Local duration evidence can then support the final decision by indicating whether the expected sound was sufficiently sustained. This choice is consistent with the strong role of recurrent and sequential models in Tajweed and Arabic pronunciation work [3, 1, 12, 10].

- **Transition-oriented rules: main transition classifier with local transition support.** Rules such as Idgham, Ikhfa', and Iqlab depend on boundary behaviour between adjacent sounds. The main classifier evaluates the broader segment, while local transition evidence supports interpretability at the boundary where the rule is expected. This design matches the fact that transition rules are neither purely static nor purely duration-based.
- **Burst-oriented rules: local event detector.** Qalqalah depends on a short, localised release-like event. A local event detector, potentially convolutional or short-context-oriented, is appropriate because the target cue is concentrated in a narrow temporal window. MFCC-based local features are suitable for this module because they preserve time-frequency detail while remaining computationally efficient [16].
- **Contextual-realisation rules: reference-aware classifier.** For rules whose correct realisation depends strongly on canonical context, the model combines aligned acoustic evidence with the rule label, textual position, and rule category supplied by the annotated reference. These cases are not purely temporal and should not be forced into the same modelling strategy as duration rules.

These choices define a coherent modelling baseline for the thesis. Each module receives the representation and model type that best matches the dominant evidence of the rules it handles. The architecture remains stable even when individual modules are refined, because routing, aggregation, and feedback operate on structured module outputs rather than on the internal details of a single model.

1.4.7 Where Temporal Modelling Becomes Necessary

Temporal modelling remains important, but it is placed in its proper conceptual location. It is not the definition of the whole modelling problem. Rather, it is required when the target rule depends on duration or on the evolution of sound across time.

- **For duration-oriented rules,** temporal modelling is essential because the judgment depends directly on how long an acoustic state persists.
- **For transition-oriented rules,** temporal modelling is also important because the judgment depends on the movement from one sound into another.

- **For burst-oriented rules**, temporal structure matters at short range, with emphasis on local event detection rather than long sequence modelling.
- **For contextual-realisation rules**, temporal modelling may support the decision, but the dominant factor is often the interaction between acoustic evidence and canonical rule context.

This explains why recurrent or sequential models remain useful inside the architecture, especially for duration and transition categories. A BiLSTM, for example, can use both preceding and following context, which is valuable in sequential recitation analysis [10]. At the same time, the architecture does not force every rule category to share the same temporal backbone. A local event detector is more appropriate for Qalqalah-like bursts, while a reference-aware classifier is more appropriate for strongly contextual realisation cases.

1.4.8 Relation to the Content Verification Model

Although this section focuses on rule-level modelling, the rule modules depend on the content verification model upstream. The content path determines whether the expected canonical segment is present and sufficiently aligned before rule analysis is applied. This separation prevents rule modules from producing misleading judgments on missing or substituted content.

Conceptually, the content model solves a different task from the rule modules. It is closer to recognition and alignment: it estimates what the learner recited and compares it with the canonical reference. For this reason, it is associated with stronger speech representations and sequence decoding, while the Tajweed rule modules are associated with category-specific acoustic evidence. The two parts are therefore complementary. The content model establishes whether the system is analysing the right segment; the specialist Tajweed modules then determine whether the expected rule was realised correctly on that segment.

1.4.9 Adaptability of the Modular Approach

A key strength of the modelling strategy is its adaptability. Because the system separates rule routing, specialist analysis, aggregation, and feedback, it does not depend on a single fixed model for all Tajweed rules. Each category can be improved or replaced without changing the overall assessment logic.

This is particularly important in Qur’anic recitation assessment, where rule categories are heterogeneous by nature. Duration-oriented rules, transition-

oriented rules, burst-oriented rules, and contextual-realisation cases place different demands on the model. A modular design therefore provides a cleaner and more principled framework than a single uniform classifier.

The architecture also supports controlled extension. A new Tajweed rule can be added by assigning it to an existing category when its behaviour matches that category, or by introducing a new specialist module when it requires a distinct analysis strategy. In both cases, the surrounding pipeline remains stable: the annotated reference supplies the rule tag, the routing function selects the responsible module, the module returns a structured judgment with evidence, and the aggregation layer incorporates that judgment into the final diagnosis.

1.5 Diagnosis Aggregation and Feedback Generation

The preceding modules produce specialised but partial outputs. The content path determines whether the learner recited the expected canonical units. The alignment process links the learner’s audio to positions in the canonical Qur’anic reference. The rule-analysis modules evaluate specific categories of Tajweed realisation, such as duration, transition, burst, or contextual behaviour. However, these outputs do not become useful to the learner until they are combined into a coherent diagnosis.

For this reason, the final stage of the architecture performs two closely related functions. First, it aggregates content-level and rule-level outputs into a structured diagnosis report. Second, it transforms that report into learner-facing feedback. The diagnosis report is the intermediate representation that connects technical model outputs to pedagogical explanation. It records what was expected, what was observed, which module produced the judgment, what evidence supported the decision, and what type of error should be communicated to the learner.

This distinction is important. The feedback layer should not generate explanations directly from raw audio. Raw audio is too low-level and ambiguous for reliable pedagogical language generation. Instead, feedback is generated from a structured diagnosis that has already localised the error, identified its type, attached it to a canonical position, and connected it to the relevant Tajweed rule when applicable.

1.5.1 Role of the Aggregation Layer

The aggregation layer receives three main streams of information.

- **Alignment information.** This identifies the correspondence between the learner’s audio and the canonical reference. It provides the position index i , the canonical unit u_i , and the time interval $[s_i, e_i]$ associated with that unit when alignment is available.
- **Content-level judgments.** These are produced by the content verification module. They indicate whether the expected canonical unit was correctly produced, substituted, deleted, or affected by insertion.
- **Rule-level judgments.** These are produced by the specialised Tajweed modules. They indicate whether an applicable rule was realised correctly and include supporting evidence such as duration estimates, transition confidence, burst evidence, or module confidence.

Figure 1.7 summarises the role of the aggregation layer in the proposed framework. The figure shows that aggregation is not a simple final classifier. Instead, it is the stage where aligned reference information, content judgments, annotated rule context, and rule-module evidence are brought together to produce a structured diagnosis report. The symbols shown on the arrows correspond to the formal elements used by the aggregation process.

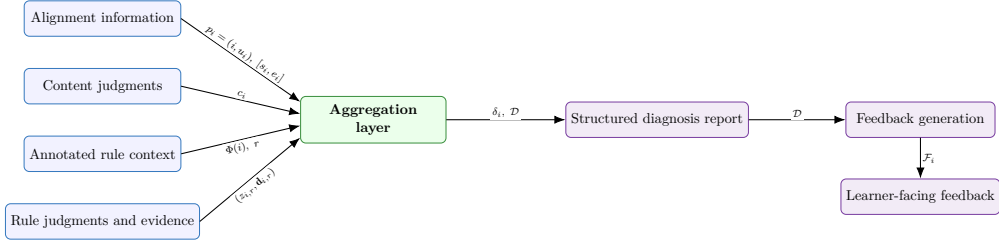


Figure 1.7: Inputs and outputs of the aggregation layer. The aggregation layer combines alignment information, content judgments, annotated rule context, and rule-level evidence into a structured diagnosis report, which is then used for feedback generation.

The aggregation layer is responsible for combining these streams without losing their meaning. A rule-level judgment is only valid when it is attached to the correct canonical position. Similarly, a content-level error must not be confused with a Tajweed-rule error. If the learner omitted or substituted the segment that should carry a rule, the primary diagnosis is a content error. If the segment is present and aligned, but the expected rule was not realised correctly, the diagnosis becomes a rule-level Tajweed error.

Thus, the aggregation layer does not simply concatenate module outputs. It applies a decision policy that preserves the logical relation between content verification, rule applicability, specialist module evidence, and final diagnosis.

1.5.2 Inputs to the Diagnosis Report

For each canonical position i , the aggregation layer considers the following information:

$$u_i, \quad [s_i, e_i], \quad c_i, \quad \Phi(i), \quad \{(z_{i,r}, \mathbf{d}_{i,r}) \mid r \in \Phi(i)\}. \quad (1.33)$$

Here, u_i is the canonical unit, $[s_i, e_i]$ is the aligned audio interval, c_i is the content judgment, $\Phi(i)$ is the set of Tajweed rules expected at that position, $z_{i,r}$ is the judgment for rule r , and $\mathbf{d}_{i,r}$ is the diagnostic evidence returned by the relevant specialist module.

The annotated reference contributes the symbolic rule information needed to interpret the acoustic result. It tells the system which rule is expected at a given position and which module category is responsible for checking it. The model output then tells the system whether the learner’s realisation matches that expectation. The diagnosis report is produced by combining these two sources: the reference defines the expected rule context, and the model provides the observed acoustic judgment.

1.5.3 Content-Priority Aggregation Logic

The aggregation stage follows a content-priority decision policy. This means that content correctness is evaluated before rule correctness. The reason is that Tajweed-rule analysis is meaningful only when the underlying canonical segment has been produced and aligned reliably.

For each canonical position i , the aggregation logic can be expressed as follows:

1. Retrieve the canonical unit u_i and its aligned interval $[s_i, e_i]$.
2. Retrieve the content judgment c_i from the content verification module.
3. If $c_i \neq \text{correct}$, create a content-level diagnosis entry and mark all rule judgments at that position as not evaluated.
4. If $c_i = \text{correct}$, retrieve the applicable rule set $\Phi(i)$ from the annotated reference.
5. For each rule $r \in \Phi(i)$, retrieve the corresponding module output $(z_{i,r}, \mathbf{d}_{i,r})$.
6. Create a rule-level diagnosis entry when $z_{i,r} = \text{incorrect}$.
7. Preserve correct rule judgments as evidence when needed, but prioritise incorrect or uncertain entries for learner-facing feedback.
8. Combine all entries into a structured report ordered by canonical position.

This logic prevents contradictory interpretations. For example, if a Madd-bearing segment is missing, the report should not tell the learner that the Madd was too short. The more accurate diagnosis is that the expected segment itself was not produced. By contrast, if the segment is present but its duration evidence is insufficient, then the report can identify a duration-related Tajweed error.

The same policy can be represented formally as

$$\delta_i = \begin{cases} \delta_i^{\text{content}}, & \text{if } c_i \neq \text{correct}, \\ \delta_i^{\text{rule}}, & \text{if } c_i = \text{correct and } \exists r \in \Phi(i) \text{ such that } z_{i,r} = \text{incorrect}, \\ \delta_i^{\text{correct}}, & \text{otherwise.} \end{cases} \quad (1.34)$$

Here, δ_i denotes the final diagnosis decision associated with position i . This decision may represent a content error, a rule-level Tajweed error, or a correct realisation.

1.5.4 Structure of the Diagnosis Report

The output of the aggregation layer is a structured diagnosis report rather than a single correctness score. Conceptually, the full diagnosis report is represented as

$$\mathcal{D} = \{\delta_1, \delta_2, \dots, \delta_N\}, \quad (1.35)$$

where each δ_i is a position-level diagnosis entry. A diagnosis entry can be defined as

$$\delta_i = (i, u_i, [s_i, e_i], c_i, \Phi(i), \mathcal{M}_i, \mathcal{E}_i, \ell_i), \quad (1.36)$$

where i is the canonical position, u_i is the canonical unit, $[s_i, e_i]$ is the aligned time interval, c_i is the content judgment, $\Phi(i)$ is the applicable rule set, $\mathcal{M}_i$ is the module or modules involved in the decision, $\mathcal{E}_i$ is the diagnostic evidence, and ℓ_i is the final diagnosis label assigned to that position.

This abstract representation can be implemented in a machine-readable format such as JSON. A simplified diagnosis entry may contain fields such as those shown in Table 1.5.

Table 1.5: Main fields of a structured diagnosis report entry.

Field	Role in the diagnosis report
Position	Identifies the canonical character, phoneme, word, or segment to which the diagnosis is attached.
Canonical unit	Stores the expected unit from the Qur’anic reference.
Aligned interval	Indicates the time span in the learner’s audio associated with the canonical unit.
Content judgment	Specifies whether the unit was correct, substituted, deleted, or affected by insertion.
Expected rule	Stores the Tajweed rule or rules expected at that position, according to the annotated reference.
Source module	Indicates which module produced the rule judgment, such as content, duration, transition, burst, or contextual analysis.
Rule judgment	Indicates whether the expected Tajweed rule was realised correctly, incorrectly, or not evaluated.
Evidence	Stores supporting information such as confidence, measured duration, transition score, burst indicator, or alignment quality.
Diagnosis label	Gives the final technical interpretation, such as content substitution, deletion, insufficient duration, weak transition, or missing release.
Feedback priority	Indicates whether the entry should be shown to the learner immediately, grouped with similar errors, or kept as internal diagnostic evidence.

This report structure is one of the main advantages of the modular and aligned approach. Because each error is linked to a canonical position and to a source module, the system can identify not only that an error occurred, but also what type of error it is. For example, an error coming from the duration module is already interpreted as a temporal Tajweed issue. When this module-level information is combined with the annotated rule label at the aligned position, the system can distinguish between a Madd-related duration problem and a Ghunnah-related duration problem. This produces a more informative diagnosis than a global correct/incorrect classifier.

1.5.5 Modular Evidence and Error Interpretation

Each specialist module contributes a different type of diagnostic evidence. The aggregation layer preserves this evidence so that the final diagnosis remains explainable.

- **Content module evidence.** This may include the predicted content sequence, edit distance, character-level mismatch, or confidence in the alignment. It supports diagnoses such as deletion, insertion, and substitution.
- **Duration module evidence.** This may include duration-related scores or confidence values. It supports diagnoses involving insufficient or excessive elongation and sustained nasalisation.
- **Transition module evidence.** This may include confidence in the expected boundary behaviour. It supports diagnoses involving assimilation, concealment, or conversion between neighbouring sounds.
- **Burst module evidence.** This may include local event confidence or evidence of a release cue. It supports diagnoses involving Qalqalah-like phenomena.
- **Contextual module evidence.** This may combine acoustic evidence with rule labels and canonical context. It supports diagnoses where the expected realisation depends on whether a sound should be pronounced, merged, reduced, or suppressed.

The source module therefore narrows the interpretation of the error. This is a direct diagnostic advantage of the modular architecture. A monolithic classifier may return that a segment is incorrect, but it does not naturally explain whether the error is textual, temporal, transitional, or release-related. The modular system, by contrast, carries this information through the pipeline because each module is associated with a specific error family.

1.5.6 From Diagnosis Report to Pedagogical Feedback

A central limitation in many automatic pronunciation and recitation systems is that they stop at detection. They may identify whether the learner is correct or incorrect, but they often do not explain how the learner should improve. This limitation is particularly important in Tajweed learning, where the pedagogical value of the system depends on identifying the relevant rule and the nature of the required correction [6, 18].

In this framework, feedback generation is based on the structured diagnosis report. The report already contains the key information needed for explanation:

- where the deviation occurred;
- whether the deviation is content-level or rule-level;
- which module produced the error signal;
- which Tajweed rule was expected at that position;
- what evidence supports the decision.

The feedback module therefore does not infer feedback directly from the waveform. It converts a structured technical diagnosis into a learner-facing message. This makes the feedback more stable, more explainable, and easier to control.

1.5.7 Feedback Types

The system distinguishes between content feedback and rule feedback.

Content feedback. Content feedback is produced when the learner does not recite the expected canonical unit. In this case, the feedback focuses on *what* was recited. For example, it may indicate that a unit was omitted, substituted, or inserted. Rule-level feedback is not prioritised for that position because the underlying segment was not reliably produced.

Rule feedback. Rule feedback is produced when the content is present and aligned, but the expected Tajweed rule was not realised correctly. In this case, the feedback focuses on *how* the unit was recited. For example, a duration-module error can be verbalised as an insufficient realisation of a Madd or Ghunnah rule, while a transition-module error can be verbalised as an incorrect boundary realisation involving Ikhfa', Idgham, or Iqlab.

This distinction is pedagogically important. The learner must know whether the problem lies in the content of the recitation or in the manner of recitation. A content error requires the learner to correct the recited unit itself. A rule error requires the learner to correct the performance of a specific Tajweed rule on an otherwise valid unit.

1.5.8 Example of Diagnosis-to-Feedback Transformation

The following abstract example illustrates how a structured diagnosis entry can be transformed into feedback.

Listing 1.1: Simplified example of a diagnosis report entry.

```
{
  "position": 12,
  "canonical_unit": "",
  "aligned_interval": [1.42, 1.76],
  "content_judgment": "correct",
  "expected_rule": "ghunnah",
  "source_module": "duration",
  "rule_judgment": "incorrect",
  "evidence": {
    "confidence": 0.87,
    "detail": "duration shorter than expected"
  },
  "diagnosis_label": "insufficient duration"
}
```

From this entry, the system can generate a technical diagnosis such as:

At the aligned position containing $\dot{\text{u}}$, the expected rule is Ghunnah. The duration module judged the realisation as insufficient.

The learner-facing feedback can then be expressed more simply:

The Ghunnah at this position was too short. Sustain the nasal sound for longer on the next recitation.

The important point is that the feedback is not guessed directly from the audio. It is derived from the structured report. The aligned position identifies where the issue occurred, the annotated reference identifies the expected rule, the source module identifies the error family, and the evidence explains why the decision was made.

1.5.9 Position of Feedback in the Overall Architecture

Within the full conceptual framework, feedback generation is the final transformation of the diagnosis report into a form that is usable for learning. It does not replace the analytical stages that precede it. Instead, it depends

on their outputs. The quality of the feedback is therefore determined by the quality of alignment, content verification, rule-specific modelling, and aggregation.

At the same time, the presence of a dedicated feedback layer gives the whole architecture a stronger pedagogical orientation. The system is not designed only to classify recitations. It is designed to support correction. By separating model outputs, structured diagnosis, and learner-facing feedback, the framework preserves both technical rigour and educational usefulness.

This design also leaves room for future improvement. The current annotated reference provides the rule labels and positional information needed for diagnosis. A later extension could enrich this reference with rule explanation metadata, such as expected realisation, correction hints, or learner-facing templates. Such an extension would make feedback generation more detailed while preserving the same modular and aligned diagnosis pipeline.

1.6 Chapter Summary

This chapter presented the conceptual framework adopted in this thesis for automatic Tajweed assessment of Qur’anic recitation. The main argument was that Tajweed assessment cannot be reduced to a single monolithic classification problem, because recitation errors differ in their source, acoustic behaviour, and pedagogical meaning. For this reason, the proposed framework separates content verification from rule-level Tajweed analysis and organises the system around specialised but cooperating modules.

The chapter first formalised the assessment problem by defining the canonical recitation sequence, the learner’s acoustic input, the annotated rule reference, the alignment relation, content judgments, rule judgments, and the final diagnosis object. This formalisation clarified what the system receives, what it knows, what it decides, and what it must return to the learner.

The architecture was then described as a modular and reference-conditioned pipeline. The annotated Qur’anic reference plays a central role by indicating which Tajweed rules apply at each canonical position. After content verification and alignment, rule-bearing segments are routed to specialist modules according to the acoustic nature of the expected rule, such as duration, transition, burst, or contextual realisation.

The chapter also justified the feature and modelling choices used in the framework. MFCC-based representations are suitable for local, temporal, and rule-oriented acoustic analysis, while self-supervised speech representations are more appropriate for content verification and alignment. Similarly, the modelling strategy follows rule categories rather than imposing one uniform model on all Tajweed phenomena.

Finally, the chapter explained how the outputs of the different modules are combined into a structured diagnosis report and then transformed into learner-facing feedback. This diagnosis-oriented design is central to the contribution of the framework: the system is not intended only to detect that an error occurred, but to identify where it occurred, whether it is a content or rule-level error, which Tajweed rule is involved, and what kind of correction the learner needs.

Bibliography

- [1] Abdelfatah Ahmed, Mohamed Bader, Ismail Shahin, Ali Bou Nassif, Naoufel Werghi, and Mohammad Basel. Arabic mispronunciation recognition system using LSTM network. *Information*, 14(7):413, 2023. doi: 10.3390/info14070413.
- [2] Mahmoud Al-Ayyoub, Nour Alhuda Damer, and Ismail Hmeidi. Using deep learning for automatically determining correct application of basic Quranic recitation rules. *International Arab Journal of Information Technology*, 15(3A):620–625, 2018.
- [3] Ahmad Al Harere and Khlood Al Jallad. Mispronunciation detection of basic Quranic recitation rules using deep learning. *arXiv preprint arXiv:2305.06429*, 2023. URL <https://arxiv.org/abs/2305.06429>.
- [4] Norah Alrashoudi, Hend Al-Khalifa, and Yousef Alotaibi. Improving mispronunciation detection and diagnosis for non-native learners of the arabic language. *Discover Computing*, 28(1):1, 2025. doi: 10.1007/s10791-024-09489-8.
- [5] Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems*, 33: 12449–12460, 2020.
- [6] Sener Birgun. A review of automated Quranic recitation assessment systems. *Journal of Islamic Studies and Technology*, 2026.
- [7] Şükrü Selim Çalık, Ayhan Küçükmanisa, and Zeynep Hilal Kilimci. A novel framework for mispronunciation detection of arabic phonemes using audio-oriented transformer models. *Applied Acoustics*, 215:109711, 2024. doi: 10.1016/j.apacoust.2023.109711.
- [8] Steven Davis and Paul Mermelstein. Comparison of parametric representations for monosyllabic word recognition in continuously spoken

- sentences. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 28(4):357–366, 1980. doi: 10.1109/TASSP.1980.1163420.
- [9] Yassine El Kheir, Omnia Ibrahim, Amit Meghanani, Nada Almarwani, Hawau Olamide Toyin, Sadeen Alharbi, Modar Alfadly, Lamya Alkanhal, Ibrahim Selim, Shehab Elbatal, Salima Mdhaffar, Thomas Hain, Yasser Hifny, Mostafa Shahin, and Ahmed Ali. Towards a unified benchmark for arabic pronunciation assessment: Qur’anic recitation as case study, 2025. Accepted at Interspeech 2025 and ArabicNLP Shared Task 2025.
 - [10] Alex Graves and Jürgen Schmidhuber. Framewise phoneme classification with bidirectional LSTM and other neural network architectures. *Neural Networks*, 18(5–6):602–610, 2005. doi: 10.1016/j.neunet.2005.06.042.
 - [11] Esraa Hassan, Abeer Saber, Omar Alqahtani, Nora El-Rashidy, and Samar Elbedwehy. An innovative approach to advanced voice classification of sacred Quranic recitations through multimodal fusion. *Egyptian Informatics Journal*, 30:100640, 2025. doi: 10.1016/j.eij.2025.100640.
 - [12] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.
 - [13] Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 706–710. IEEE, 2021. doi: 10.1109/ICASSP39728.2021.9414231.
 - [14] Ahmed Ibrahim. Hafs2Vec: A system for the IqraEval Arabic and Qur’anic phoneme-level pronunciation assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 453–456, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.arabnlp-sharedtasks.62.
 - [15] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *Proceedings of the International Conference on Machine Learning*, pages 28492–28518, 2023.
 - [16] Mas Muhammad Aqil Salim, Ani Dijah Rahajoe, and Anggraini Puspita Sari. Dataset ratio experiment for Tajweed law recognition using MFCC

- and CNN features. *JEEM ECS*, 8(2):91–97, 2025. doi: 10.26905/jeemecs.v8i2.14789.
- [17] Vibha Tiwari. MFCC and its applications in speaker recognition. *International Journal on Emerging Technologies*, 1(1):19–22, 2010.
- [18] Yanlu Xie, Huihang Zhong, Xuhui Lan, and Wenwei Dong. Mispronunciation detection and diagnosis based on large language models. *Computer Speech & Language*, 99:101942, 2026. doi: 10.1016/j.csl.2026.101942.